

```
(%i4) /* Symbols and functions */
declare(x, real);
declare([x_n, h, v_n], [real, constant]);
assume(h > 0);
depends([y, v], x);
```

```
(%o1) done
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```
(%o2) done
```

```
(%o3) [0 < h]
```

```
(%o4) [y(x), v(x)]
```

```
(%i6) /* Taylor expansion of y(x) around x_n */
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```
taylor(y(x), x, x_n, 4);$;
```

```
y_plus: subst(x - x_n = h, %);
```

$$y_{\text{plus}} = \frac{h^4 \left(\frac{d^4}{dx^4} y(x) \right) \Big|_{x=x_n}}{24} + \frac{h^3 \left(\frac{d^3}{dx^3} y(x) \right) \Big|_{x=x_n}}{6} + \frac{h^2 \left(\frac{d^2}{dx^2} y(x) \right) \Big|_{x=x_n}}{2} + h \left(\frac{d}{dx} y(x) \right) \Big|_{x=x_n} + y(x_n)$$

```
(%i7) /* Apply the system: y'(x) = v(x), v'(x) = x y(x) */
```

```
y_plus: subst(
```

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[
```

```
at(diff(y(x), x, 4), x = x_n) = at(diff(x · y(x), x, 2), x = x_n),
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at(diff(y(x), x, 3), x = x_n) = at(diff(x · y(x), x, 1), x = x_n),
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```
at(diff(y(x), x, 2), x = x_n) = at(x · y(x), x = x_n),
```

```
at(diff(y(x), x, 1), x = x_n) = at(v(x), x = x_n)
```

```
], y_plus), nouns; /* y_{n+1} = ... */
```

$$y_{\text{plus}} = \frac{h^4 (x_n^2 y(x_n) + 2 v(x_n))}{24} + \frac{h^3 (y(x_n) + x_n v(x_n))}{6} + \frac{h^2 x_n y(x_n)}{2} + y(x_n) + h v(x_n)$$

```
(%i9) /* Taylor expansion of v(x) = y'(x) around x_n */
```

```
taylor(v(x), x, x_n, 4);$;
```

```
v_plus: subst(x - x_n = h, %);
```

$$v_{\text{plus}} = \frac{h^4 \left(\frac{d^4}{dx^4} v(x) \right) \Big|_{x=x_n}}{24} + \frac{h^3 \left(\frac{d^3}{dx^3} v(x) \right) \Big|_{x=x_n}}{6} + \frac{h^2 \left(\frac{d^2}{dx^2} v(x) \right) \Big|_{x=x_n}}{2} + h \left(\frac{d}{dx} v(x) \right) \Big|_{x=x_n} + v(x_n)$$

(%i10) /* Apply the system: $y'(x) = v(x)$, $v'(x) = x y(x)$ */

v_plus: subst(

[

at(diff(v(x), x, 4), x = x_n) = at(diff(x · y(x), x, 3), x = x_n),
at(diff(v(x), x, 3), x = x_n) = at(diff(x · y(x), x, 2), x = x_n),
at(diff(v(x), x, 2), x = x_n) = at(diff(x · y(x), x, 1), x = x_n),
at(diff(v(x), x, 1), x = x_n) = at(x · y(x), x = x_n),
at(diff(y(x), x, 4), x = x_n) = at(diff(x · y(x), x, 2), x = x_n),
at(diff(y(x), x, 3), x = x_n) = at(diff(x · y(x), x, 1), x = x_n),
at(diff(y(x), x, 2), x = x_n) = at(x · y(x), x = x_n),
at(diff(y(x), x, 1), x = x_n) = at(v(x), x = x_n)

], v_plus), nouns; /* v_{n + 1} = ... */

v_plus
$$\frac{h^4 (x_n^2 (y(x_n) + x_n v(x_n)) + 3 x_n y(x_n))}{24} + \frac{h^3 (x_n^2 y(x_n) + 2 v(x_n))}{6} + \frac{h^2 (y(x_n) + x_n v(x_n))}{2} + h x_n$$

$$y(x_n) + v(x_n)$$

(%i12) /* Results */

map(factor, collectterms(y_plus, h));

map(factor, collectterms(v_plus, h));

(%o11)
$$\frac{h^4 (x_n^2 y(x_n) + 2 v(x_n))}{24} + \frac{h^3 (y(x_n) + x_n v(x_n))}{6} + \frac{h^2 x_n y(x_n)}{2} + y(x_n) + h v(x_n)$$

(%o12)
$$\frac{h^3 (x_n^2 y(x_n) + 2 v(x_n))}{6} + \frac{h^4 x_n (4 y(x_n) + x_n v(x_n))}{24} + \frac{h^2 (y(x_n) + x_n v(x_n))}{2} + h x_n y(x_n) + v(x_n)$$